

if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Monitor weight and BMI during treatment

* Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic

- Should check blood pressure in the elderly before starting and for the first few weeks
- Monitoring elevated prolactin levels of dubious clinical benefit
- Phenothiazines may cause false-positive phenylketonuria results
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and

pipothiazine should be discontinued at the first sign of decline of WBC in the absence of other causative factors

- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope

- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown

Notable Side Effects

- * Excitement, insomnia, restlessness
- * Tardive dyskinesia (risk increases with duration of treatment and with dose)
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Galactorrhea, amenorrhea
- Dry mouth, nausea, blurred vision, sweating, appetite change
- Sexual dysfunction (impotence)
- Hypotension, arrhythmia, tachycardia
- Weight gain
- Rare rash



Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Jaundice, leukopenia

- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Many experience and/or can be significant in amount

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose
- For sedation, take at night
- Switch to an atypical antipsychotic

- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 50–100 mg once a month

Dosage Forms

- Injection 50 mg/mL

How to Dose

- Initial 25 mg; can be increased by 25–50 mg; maximum 200 mg once a month

- Drug should be administered intramuscularly in the gluteal region



Dosing Tips

* Only available as long-acting intramuscular formulation and not as oral formulation

- The peak of action generally occurs after 9–10 days
- May need to treat with an oral antipsychotic for 1–2 weeks when initiating treatment
- One of the few conventional antipsychotics available in a depot formulation lasting for up to a month
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Sedation, tachycardia, drug-induced parkinsonism, arrhythmia, hypothermia, ECG changes, hypotension

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- If antiparkinson agents are being used, they should be continued for a few weeks after pipothiazine is discontinued

Pharmacokinetics

- Onset of action of palmitic ester formulation within 2–3 days
- Duration of action of palmitic ester formulation 3–6 weeks



Drug Interactions

- Use with TCAs may increase risk of cardiac symptoms
- CNS effects may be increased if used with other CNS depressants
- May increase the effects of antihypertensive agents
- May decrease the effects of amphetamines, levodopa, dopamine agonists, clonidine, guanethidine, adrenaline
- Effects may be reduced by anticholinergic agents
- Antacids, antiparkinson drugs, and lithium may reduce pipothiazine absorption
- Combined use with epinephrine may lower blood pressure
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome



Other Warnings/Precautions

- Patients may be more sensitive to extreme temperatures
- Use with caution in patients with Parkinson's disease, Lewy body dementia, or drug-induced parkinsonism with previous treatments
- Avoid undue exposure to sunlight
- Use with caution in patients with respiratory disease, Lewy body dementia, angle-closure glaucoma (including family history), alcohol withdrawal syndrome, brain damage, epilepsy, hypothyroidism, myaesthesia gravis, prostatic hypertrophy, thyrotoxicosis
- Contact with skin can cause rash
- Antiemetic effect can mask signs of other disorders or overdose
- Do not use epinephrine in event of overdose as interaction with some pressor agents may lower blood pressure

Do Not Use

- If patient is comatose
- If there is cerebral atherosclerosis
- If patient has phaeochromocytoma
- If patient has renal or liver failure, blood dyscrasias
- If patient has severe cardiac impairment
- If patient has subcortical brain damage
- If there is a proven allergy to pipothiazine

- If there is a known sensitivity to any phenothiazine

Special Populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Use with caution

Elderly

- Elderly patients do not metabolize the drug as quickly
- Dose should be reduced
- Recommended starting dose 5–10 mg
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

Children and Adolescents

- Not recommended for use in children

Pregnancy

- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Reports of drug-induced parkinsonism, jaundice, hyperreflexia, hyporeflexia in infants whose mothers took a phenothiazine during pregnancy
- Not recommended unless absolutely necessary
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

Breast Feeding

- Unknown if pipothiazine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Noncompliant patients

Potential Disadvantages

- Patients who need immediate onset of antipsychotic actions

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Aggressive symptoms

Pearls

- Pipothiazine is a higher potency phenothiazine
- Less risk of sedation and orthostatic hypotension but greater risk of drug-induced parkinsonism than with low-potency phenothiazines
- * Only available in long-acting parenteral formulation
- Generally, patients must be stabilized on an oral antipsychotic prior to switching to parenteral long-acting pipothiazine
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients

can occasionally vary greatly from one atypical antipsychotic to another

- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional antipsychotic such as pipothiazine or from switching to a conventional antipsychotic such as pipothiazine
- However, long-term polypharmacy with a combination of a conventional antipsychotic with an atypical antipsychotic such as pipothiazine may combine their side effects without clearly augmenting the efficacy of either
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic

Suggested Reading

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Pitolisant

Therapeutics

Brands

- Wakix
- *see index for additional brand names*

Generic?

No

Class

- Histamine 3 antagonist/inverse agonist

Commonly Prescribed for

(bold for FDA approved)

- **Reducing excessive daytime sleepiness in patients with narcolepsy**
- **Cataplexy**

How the Drug Works

- Histamine is a wake-promoting neurotransmitter released from the tuberomammillary nucleus in the hypothalamus

- Histamine 3 receptors are presynaptic autoreceptors; thus, when histamine binds to these receptors it shuts down further histamine release
- Antagonism/inverse agonism of histamine 3 receptors by pitolisant therefore disinhibits histamine release, boosting histamine levels in the brain and enhancing wakefulness

How Long Until It Works

- Can take up to 8 weeks for some patients to achieve a clinical response

If It Works

- Improves daytime sleepiness and may improve fatigue
- May improve cataplexy
- Treat until improvement stabilizes and then continue treatment indefinitely as long as improvement persists

If It Doesn't Work

- Augment or consider an alternative treatment



Best Augmenting Combos for Partial Response or Treatment Resistance

- Stimulants
- Modafinil or armodafinil

- Best to attempt another monotherapy prior to augmenting with other drugs in the treatment of sleepiness associated with sleep disorders

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Increased histamine in the brain can lead to insomnia or anxiety, decreased appetite, and nausea

Notable Side Effects

- Insomnia, anxiety, nausea, decreased appetite



Life-Threatening or Dangerous Side Effects

- Increased heart rate, tachycardia
- Visual hallucinations, hypnagogic hallucinations

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Lower the dose
- If unacceptable side effects persist, discontinue use

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 17.8–35.6 mg once daily upon wakening

Dosage Forms

- Tablet 4.45 mg, 17.8 mg

How to Dose

- Initial dose 8.9 mg once daily; after 1 week increase to 17.8 mg once daily; after 2 weeks can increase to 35.6 mg once daily if needed; maximum dose 35.6 mg once daily



Dosing Tips

- 4.45 mg tablet contains 5 mg of pitolisant hydrochloride equivalent to 4.45 mg of pitolisant; 17.8 mg tablet contains 20 mg of pitolisant hydrochloride equivalent to 17.8 mg of pitolisant

Overdose

- Limited data

Long-Term Use

- Has been evaluated and found safe and effective in trials up to 1 year
- The need for continued treatment should be reevaluated periodically

Habit Forming

- No

How to Stop

- Taper may be prudent

Pharmacokinetics

- Extensively metabolized by CYP450 2D6 and to a lesser extent by CYP450 3A4
- No active metabolites
- Median half-life approximately 20 hours

- Food does not affect absorption

Drug Interactions

- In patients taking strong CYP450 2D6 inhibitors the maximum dose is 17.8 mg once daily
- For patients on a stable dose of pitolisant, reduce the pitolisant dose by half upon initiating a strong CYP450 2D6 inhibitor
- In patients who are poor metabolizers of CYP450 2D6 the maximum dose is 17.8 mg once daily
- Strong CYP450 3A4 inducers may reduce exposure of pitolisant
- For patients on a stable dose of pitolisant, assess for loss of efficacy after initiating a strong CYP450 3A4 inducer and consider dose adjustment
- If concomitant use of a strong CYP450 3A4 inducer is discontinued, decrease pitolisant dose by half
- Due to weak induction of CYP450 3A4, pitolisant may reduce the effectiveness of sensitive CYP450 3A4 substrates, including hormonal contraceptives; patients should use an alternative nonhormonal contraceptive method during treatment with pitolisant and for at least 21 days after discontinuation of treatment
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval
- Histamine 1 receptor antagonists may reduce effectiveness of pitolisant

Other Warnings/Precautions

- Pitolisant prolongs the QT interval; its use should be avoided in patients with known QT prolongation, in combination with other drugs known to prolong QT interval, in patients with a history of cardiac arrhythmias, and in patients with symptomatic bradycardia, hypokalemia, or hypomagnesemia
- Risk of QT prolongation may be greater in patients with hepatic or renal impairment due to higher concentrations of pitolisant; such patients should be monitored for increased QTc
- The effectiveness of hormonal contraceptives may be reduced when used with pitolisant and for 21 days after discontinuation of pitolisant
- Pitolisant is not a replacement for sleep

Do Not Use

- If patient has severe hepatic impairment
- If there is a proven allergy to pitolisant

Special Populations

Renal Impairment

- Pitolisant prolongs the QT interval, so monitor patients with renal impairment for increased QTc

- Moderate to severe renal impairment: initial dose 8.9 mg once daily; titrate to a maximum dose of 17.8 mg once daily after 7 days
- Not recommended for use in end-stage renal disease

Hepatic Impairment

- Dose adjustment not necessary in patients with mild hepatic impairment
- Moderate hepatic impairment: initial dose 8.9 mg once daily; titrate to a maximum dose of 17.8 mg once daily after 14 days
- Pitolisant prolongs the QT interval, so monitor patients with hepatic impairment for increased QTc
- Severe hepatic impairment: contraindicated

Cardiac Impairment

- Pitolisant prolongs the QT interval, so its use should be avoided in patients with a history of cardiac arrhythmias, symptomatic bradycardia, hypokalemia, hypomagnesemia, or congenital prolongation of the QT interval

Elderly

- Limited experience in patients over 65
- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Case reports have not determined a drug-associated risk of major birth defects, miscarriage, or adverse maternal or fetal outcomes
- When administered to pregnant rats during organogenesis, the no observed adverse effect level (NOAEL) for embryofetal toxicity is 27 times the maximum recommended human dose (MRHD) based on mg/m^2 body surface area
- When administered to pregnant rabbits during organogenesis, the NOAELs for maternal toxicity and embryofetal development are 2 and 4 times the MRHD based on mg/m^2 body surface
- When administered to pregnant rats from gestation day 7 through lactation, maternal toxicity, including death, occurred at 22 times the MRHD based on mg/m^2 body surface area; at the maternally toxic dose, fetal toxicity and major malformations occurred
- Pregnancy registry for pitolisant: 1-800-833-7460

Breast Feeding

- Unknown if pitolisant is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Minimal abuse potential

Potential Disadvantages

- Did not demonstrate non-inferiority to modafinil in clinical trials

Primary Target Symptoms

- Daytime sleepiness

Pearls

- Pitolisant has been approved in the EU since 2016 for the treatment of narcolepsy with or without cataplexy in adults
- Pitolisant is the only anti-narcoleptic drug that is not scheduled as a controlled substance in the USA
- Pitolisant was formerly known as tiprolisant

- Preliminary controlled data have shown that pitolisant may reduce daytime sleepiness in obstructive sleep apnea

Suggested Reading

Dauvilliers Y , Arnulf I , Szakacs Z , et al. Long-term use of pitolisant to treat patients with narcolepsy: Harmony III Study . *Sleep* 2019 ;**42** (11). pii: zsz174 . doi: 10.1093/sleep/zsz174 .

Dauvilliers Y , Verbraecken J , Partinen M , et al. Pitolisant for daytime sleepiness in obstructive sleep apnea patients refusing CPAP: a randomized trial . *Am J Respir Crit Care Med* 2020 ;**201** (9):1135–45 . doi: 10.1164/rccm.201907-1284OC .

Kollb-Sielecka M , Demolis P , Emmerich J , et al. The European Medicines Agency review of pitolisant for treatment of narcolepsy: summary of the scientific assessment by the Committee for Medicinal Products for Human Use . *Sleep Med* 2017 ;**33** :125–9 .

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